

Module specification

Title	Programming 02 – Programming for the Web				
Code	COM4012M	School	Science Technology & Health	Cost centre	2511
Level	4	Credits	20	Available for incoming study abroad	No
Pre-requisites¹				Barred combinations	None
Total number of study hours for the module:				200	
which will include the following:				Number of hours:	
timetabled contact				48	
placement				0	
field trips				0	
other - please give further detail below:					
All activities in this module relate to transferable skills, employability and gaining network development skills. Students will have opportunities to develop skills such as • Web development and scripting • Problem solving • Time management • Analytical skills and achieve relevant professional certifications.					

Rationale
<i>Please provide a concise rationale (not content) for the module, including where it sits within the programme in terms of feeding from, into and across other modules</i>
This subject introduces students to the fundamental concepts and models of Web application development. Students will be equipped with the skills to create platform independent and responsive web applications. The technologies covered in the module include HTML, CSS, JavaScript, PHP and ASP.NET. Students will aim to create a portfolio of webpages that demonstrate a range of HTML and CSS features. Students will also aim to develop a JavaScript application in response to a specified problem and implement web security. By the end of the module students would be able to analyse the extent to which a web application meets the criteria defined for its current use and future development.

Title(s) of awards to which the module contributes	Award Programme Learning Outcome(s) to which the module is mapped (PLO4.1, PLO5.3 etc.)
BSc Cyber Security	4.1 – 4.6
BSc Cyber Security with a Year in Industry	4.1 – 4.6
BSc Computer Science	4.1 – 4.6
BSc Computer Science with a Year in Industry	4.1 – 4.6
BSc Games Development	4.1 – 4.6
BSc Games Development with a Year in Industry	4.1 – 4.6
BSc Software Engineering	4.1 – 4.6
BSc Software Engineering with a Year in Industry	4.1 – 4.6

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Assessment				
#	type	description	weighting	Exempt from anonymous marking?
1	Portfolio	Create and test a cross platform website with interactive elements using a scripting language	100%	Y

Indicative content
The indicative content of the module may include the following: • Problem solving and decision-making techniques • Programming concepts • Concepts of responsive designs and platform independence • HCI principles and universal accessibility • Cascading Style Sheet (CSS) Basics • Client-side scripting such as JS, CSS and HTML • Graphics & Text Styling Basics • Media & Interactivity Basics • PHP, PHP Framework and how PHP interacts with databases • Introduction to ASP.NET. • Processing forms with PHP • Website security: content security policies, headers, cookies etc.

Reading list
Recommended reading list • Ruby, S., Copeland, D.B. and Thomas, D., 2020. Agile Web Development with Rails 6. Pragmatic bookshelf. • Duckett, J. (2014). Web design with HTML, CSS, JavaScript and jQuery set. Wiley Publishing. • Nixon, R., 2014. Learning PHP, MySQL & JavaScript: With jQuery, CSS & HTML5. " O'Reilly Media, Inc." • Freeman, A., 2016. Pro Asp. net core MVC. Apress. Required reading list • Firtman, M., 2010. Programming the mobile web. " O'Reilly Media, Inc." • Chang, S.S., 2016. <i>Go Web Programming</i>. Simon and Schuster.

Version	1	In use from	2022/23	to	
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Date approved:	18/5/22
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